

Problem Set 12 Answers

1. We have $E(X_1^2) = (1^2)(1/6) + (2^2)(1/6) + \cdots + (6^2)(1/6) = 91/6$ and $E(X_2^2) = (1^2)(1/8) + (2^2)(1/8) + \cdots + (8^2)(1/8) = 51/2$. We already know that $E(X_1) = 7/2$ so $\text{Var}(X_1) = 91/6 - (7/2)^2 = 35/12$. Similarly, $E(X_2) = 9/2$ so $\text{Var}(X_2) = 51/2 - (9/2)^2 = 21/4$. We conclude that, since X_1 and X_2 are independent, it follows that $\text{Var}(X) = \text{Var}(X_1) + \text{Var}(X_2) = 35/12 + 21/4 = 49/6$.

2a. We have $E(X^2) = (0^2)(125/216) + (1^2)(25/72) + (2^2)(5/72) + (3^2)(1/216) = 2/3$.

2b. We have $E(X^2) = (6)E(X_1X_2) + (3)E(X_1^2)$. The X_j s are independent, so $E(X_1X_2) = E(X_1)E(X_2)$. Also we note that $E(X_1^2) = E(X_1)$. So we conclude $E(X^2) = (6)(1/6)(1/6) + (3)(1/6) = 2/3$.

2c. We have $\text{Var}(X) = E(X^2) - (E(X))^2 = 2/3 - (1/2)^2 = 5/12$.

2d. We note that $\text{Var}(X_j)$ is the same for $j = 1, 2, 3$. So we have $\text{Var}(X) = (3) \text{Var}(X_1) = (3)(E(X_1^2) - (E(X_1))^2) = (3)(1/6 - (1/6)^2) = 5/12$.

3a. We have $E(X^2) = (1^2)(3/10) + (2^2)(7/30) + (3^2)(7/40) + (4^2)(1/8) + (5^2)(1/12) + (6^2)(1/20) + (7^2)(1/40) + (8^2)(1/120) = 209/20$.

3b. We have $E(X^2) = 1 + 14E(X_1X_2) + 7E(X_1^2) + 42E(X_1X_2) = 1 + 14(1/4) + 7(1/4) + 42(2)(1/4)(1/5) = 209/20$.

3c. We conclude that $\text{Var}(X) = E(X^2) - (E(X))^2 = 209/20 - (11/4)^2 = 231/80$.

3d. The X_j s are dependent. For instance, we have $E(X_1X_2) = (2)(1/4)(1/5) = 1/10$ but $E(X_1)E(X_2) = (1/4)^2 = 1/16$.

4a. We have $E(X^2) = (0^2)(32/243) + (1^2)(80/243) + (2^2)(80/243) + (3^2)(40/243) + (4^2)(10/243) + (5^2)(1/243) = 35/9$.

4b. We have $E(X^2) = (20)(E(X_1X_2)) + (5)(E(X_1^2))$. The X_j s are independent, so $E(X_1X_2) = E(X_1)E(X_2)$. Also we note that $E(X_1^2) = E(X_1)$. So we conclude $E(X^2) = (20)(1/3)(1/3) + (5)(1/3) = 35/9$.

4c. We have $\text{Var}(X) = E(X^2) - (E(X))^2 = 35/9 - (5/3)^2 = 10/9$.

4d. We note that $\text{Var}(X_j)$ is the same for $j = 1, 2, 3, 4, 5$. So we have $\text{Var}(X) = (5) \text{Var}(X_1) = (5)(E(X_1^2) - (E(X_1))^2) = (5)(1/3 - (1/3)^2) = 10/9$.

5a. We have $E(X^2) = \sum_{x=0}^{\infty} (x^2) \left(\frac{3^x e^{-3}}{x!} \right) = \sum_{x=0}^{\infty} (x)(x-1) \left(\frac{3^x e^{-3}}{x!} \right) + \sum_{x=0}^{\infty} (x) \left(\frac{3^x e^{-3}}{x!} \right) = \sum_{x=2}^{\infty} (x)(x-1) \left(\frac{3^x e^{-3}}{x!} \right) + \sum_{x=1}^{\infty} (x) \left(\frac{3^x e^{-3}}{x!} \right) = \sum_{x=2}^{\infty} \left(\frac{3^x e^{-3}}{(x-2)!} \right) + \sum_{x=1}^{\infty} \left(\frac{3^x e^{-3}}{(x-1)!} \right)$. Now shifting indices, we have $E(X^2) = \sum_{x=0}^{\infty} \left(\frac{3^{x+2} e^{-3}}{x!} \right) + \sum_{x=0}^{\infty} \left(\frac{3^{x+1} e^{-3}}{x!} \right) = (3^2) \sum_{x=0}^{\infty} \left(\frac{3^x e^{-3}}{x!} \right) + (3) \sum_{x=0}^{\infty} \left(\frac{3^x e^{-3}}{x!} \right) = 3^2 + 3 = 12$.

5b. We have $\text{Var}(X) = E(X^2) - (E(X))^2 = 12 - 3^2 = 3$.